	GLOBAL ECONOMICS GRADE: 11TH TERM 2 – LESSON 3 PROJECT PROJECT PROBLEMS TEACHER'S NAME: Nicolás López Cuéllar	SECOND TERM 2025–2026

Learning objective: Calculate and interpret confidence intervals for the population mean using the t-student distribution, and justify interval-based decisions when variance is unknown in finance and economy contexts.	**Assessment criteria:** C8: Calculates confidence intervals applying the t-student distribution in context situations. C9: Estimates the confidence interval for the mean when the variance is unknown in context situations.

Criteria assessment

Each assessment criterion is evaluated across the ten problems. A criterion is considered passed when it is correctly activated in at least 8 of the 10 problems.

1. Driver payouts in ride-sharing

A ride-sharing platform studies daily gross driver payouts (hundreds of USD) in one urban zone. A random sample of $n = 10$ days gives the raw data:

$$42,\ 38,\ 45,\ 40,\ 41,\ 39,\ 44,\ 43,\ 37,\ 41$$

1) Construct a 95% confidence interval for the population mean using the t-distribution. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

2. Platform fees for investment app users

An investment app analyzes mean per-trade platform fees (USD) paid by users in their 20s. A random sample of $n = 18$ transactions gives the raw data:

$$1.8,\ 2.1,\ 1.9,\ 2.4,\ 2.0,\ 1.7,\ 2.2,\ 2.3,\ 1.8,$$
$$2.5,\ 2.1,\ 1.9,\ 2.0,\ 2.2,\ 1.6,\ 2.4,\ 2.3,\ 1.9$$

1) Construct a 90% confidence interval for the population mean using the t-distribution. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

3. Weekday-morning order values

A food-delivery app studies mean order value (USD) during weekday mornings for young professionals. A random sample of $n = 30$ receipts gives the raw data:

$$115,\ 122,\ 118,\ 130,\ 126,\ 119,\ 121,\ 124,$$
$$128,\ 117,\ 123,\ 120,\ 125,\ 129,\ 116,\ 127,$$
$$122,\ 118,\ 124,\ 121,\ 126,\ 119,\ 123,\ 128,$$
$$120,\ 125,\ 117,\ 124,\ 122,\ 126$$

1) Construct a 99% confidence interval for the population mean using the t-distribution. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

4. Same-city delivery time classes

A gig-delivery startup studies average delivery time (minutes) for same-city orders. For $n = 60$ orders, data are grouped as follows:

Class (minutes)	Midpoint m_i	f_i
20–24	22	6
25–29	27	9
30–34	32	14
35–39	37	13
40–44	42	10
45–49	47	8

1) Construct a 95% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

5. Daily ad spending classes

A direct-to-consumer e-commerce startup estimates mean daily ad spending (thousand USD). For

$n = 120$ days, grouped data are:

Class (thousand USD)	m_i	f_i
50–55	52.5	12
55–60	57.5	18
60–65	62.5	28
65–70	67.5	24
70–75	72.5	22
75–80	77.5	16

1) Construct a 90% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

6. Monthly recurring revenue classes

A subscription software network studies mean monthly recurring revenue (thousand USD) for small creator-led apps. For $n = 250$ firms, grouped data are:

Class (thousand USD)	m_i	f_i
80–90	85	18
90–100	95	32
100–110	105	46
110–120	115	58
120–130	125	44
130–140	135	30
140–150	145	22

1) Construct a 95% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

7. Monthly account balance classes

A fintech savings app estimates the mean monthly account balance (hundreds of USD) of active users in their 20s. For $n = 500$ users, grouped data are:

Class (hundreds of USD)	m_i	f_i
60–65	62.5	30
65–70	67.5	52
70–75	72.5	84
75–80	77.5	108
80–85	82.5	96
85–90	87.5	68
90–95	92.5	40
95–100	97.5	22

1) Construct a 99% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

8. Annual instructor earnings classes

A national online tutoring marketplace estimates mean annual instructor earnings (USD) for active tutors. For $n = 1000$ policies, grouped data are:

Class (USD)	m_i	f_i
125–135	130	70
135–145	140	130
145–155	150	220
155–165	160	250
165–175	170	180
175–185	180	100
185–195	190	50

1) Construct a 95% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

9. Monthly seller revenue classes

An e-commerce marketplace analyzes mean monthly seller revenue (thousand USD) per store.

For $n = 2500$ merchants, grouped data are:

Class (thousand USD)	m_i	f_i
200–220	210	180
220–240	230	360
240–260	250	620
260–280	270	700
280–300	290	380
300–320	310	180
320–340	330	80

1) Construct a 90% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

10. Quarterly creator revenue classes

A streaming and creator-services network studies mean quarterly creator revenue (thousand USD) across channels. For $n = 5000$ firms, grouped data are:

Class (thousand USD)	m_i	f_i
400–440	420	320
440–480	460	680
480–520	500	1300
520–560	540	1400
560–600	580	820
600–640	620	360
640–680	660	120

1) Construct a 99% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.